Reader Report: Week 06

Probabilistic Invariance: VI-BIP and the Core Theorems

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Summary

Wu and Blei (2025) place a prior over binary selectors $z \in \{0,1\}^p$, where $z_j = 1$ indexes a feature in the invariant set S^* (Peters et al., 2016). The posterior $p(z \mid \mathcal{D}) \propto p(z) \prod_{e,i} \hat{g}(Y_i^e \mid X_{ei}^z)/\hat{p}_e(Y_i^e \mid X_{ei}^z)$ rewards selectors for which the pooled conditional $\hat{g}$ dominates each environment-specific conditional $\hat{p}_e$. Consistency (Thm 1) requires $\mu_{\min} > 0$, the minimum KL divergence between $\hat{g}$ and any $\hat{p}_e$ over non-invariant features (Wu and Blei, 2025, Def. 1). Contraction (Thm 2) gives $\mathrm{TV}(p(z \mid \mathcal{D}), \delta_{z^*}) = O(2^p\, e^{-\kappa n E \mu_{\min}})$. Both bounds assume squared loss under a linear model, and the product likelihood weights all environments symmetrically. VI-BIP (Wu and Blei, 2025, Sec. 4) approximates the posterior via mean-field variational inference (Blei et al., 2017), optimizing the ELBO with the U2G gradient estimator.

Critique

Peters et al. (2016), Fan et al. (2024), and Wu and Blei (2025) all enforce the same invariance condition, yet what separates them is what the practitioner assumes about the environments before seeing data.

When the practitioner *does* know which environment carries the strongest signal—e.g., a single CRISPR knockout versus observational controls—BIP's symmetric product likelihood cannot exploit this. It weights every environment equally, so the contraction rate $nE \gtrsim O(p/\mu_{\min})$ depends on the *average* signal across environments rather than the maximum. A practitioner holding domain knowledge about intervention strength pays the same sample cost as one who holds none.

NegDRO can exploit such knowledge through negative environment weights (Fan et al., 2024, Def. 1), but no theory characterizes the boundary at which asymmetric weighting becomes preferable to symmetric. A practitioner facing a new multi-environment dataset therefore has no decision rule for choosing between the two frameworks, nor any way to verify post hoc that the choice was appropriate.

The empirical evaluation does not help resolve this. Wu and Blei (2025, Sec. 5) validate VI-BIP exclusively on CRISPR knockouts where $\mu_{\min} > 0$ holds by construction—the regime where BIP's agnosticism is least needed. The theoretical guarantee holds regardless, but the empirical case for VI-BIP in observational or low-intervention settings, where the symmetric design should matter most, remains unmade.

Questions

Both questions below concern whether the practitioner-knowledge gap between BIP and NegDRO can be bridged.

1. The critique above treats practitioner knowledge as a binary: either environments are exchangeable (ICP/BIP) or one dominant intervention is known (NegDRO). Can this be formalized as a continuous parameter, so that a single framework interpolates between symmetric and asymmetric environment weighting? If so, does the minimax estimation rate vary smoothly along this axis, or is there a phase transition at which asymmetric weighting becomes strictly preferable?
2. Nonparametric extensions of ICP exist for the symmetric case (Heinze-Deml et al., 2018), but no nonparametric analogue of NegDRO's asymmetric weighting has been proposed. Does the practitioner-knowledge distinction identified above survive when the loss is no longer quadratic, or does the symmetry-versus-asymmetry tradeoff collapse under nonparametric conditionals?

References

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